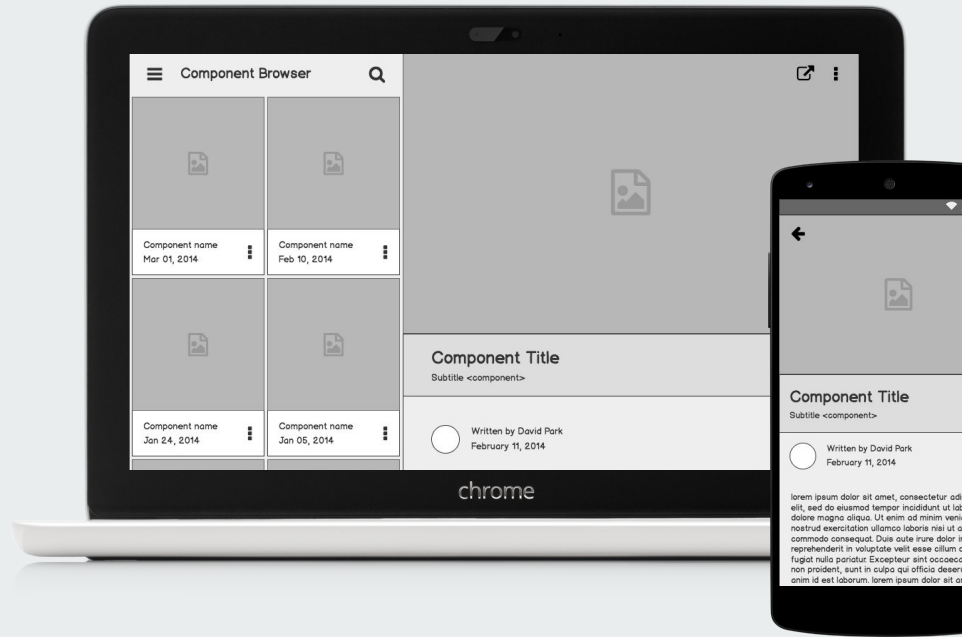




# Project Capstone



Andrea Morales Salcedo  
DEFT JUL2026

---

# Phase 1

```
-- Module 3: Fraud Pattern Analysis
-- Q11. Calculate fraud rate by:
--      Merchant Category

SELECT
    merchant_category,
    ROUND(COUNT_IF(is_fraud = TRUE)
    / COUNT(transaction_id) * 100.0, 2) AS fraud_rate
FROM bank_fraud
GROUP BY merchant_category
ORDER BY fraud_rate DESC;

--- Q12. Determine fraud rate by:
--- Payment Method

SELECT
    payment_method,
    ROUND(COUNT_IF(is_fraud = TRUE)
    / COUNT(transaction_id) * 100.0, 2) AS fraud_rate
FROM bank_fraud
GROUP BY payment_method
ORDER BY fraud_rate DESC;

-- Q13. Determine fraud rate by:
-- Device Type
```

Dataset : Bank Fraud dataset

Use SQL queries in Snowflake to create tables/databases and answer the following questions.

Module 1: Data Exploration & Profiling

Module 2: Customer Behavior Analytics

Module 3: Fraud Pattern Analysis

# Phase 2

```
Module 1: Data Profiling & Quality

Q1. Create a complete Data Quality Report showing data type, null count, null percentage, distinct count, duplicate count, and min/max values where applicable.

• Objective: Solve the following advanced-level Python Pandas and Data Analytics exercises using the Telecom Customer Churn dataset.

import pandas as pd

df = pd.read_csv("/Users/andreams/capstone_project/datasets/telecom_customers_churn.csv")

df.shape

(7043, 21)

# DATA QUALITY REPORT
data_quality_report = pd.DataFrame()

data_quality_report["data_type"] = df.dtypes
data_quality_report["null_count"] = df.isnull().sum()
data_quality_report["null_percentage"] = (df.isnull().mean() * 100).round(2)
data_quality_report["distinct_count"] = df.nunique()
data_quality_report["duplicate_count_rows"] = df.duplicated().sum()
data_quality_report["duplicate_value_count"] = df.apply(lambda x: x.duplicated().sum(), axis=1).sum()

data_quality_report["min_value"] = df.select_dtypes(include="number").min()
data_quality_report["max_value"] = df.select_dtypes(include="number").max()

data_quality_report
```

## Telecom Customer Churn Dataset

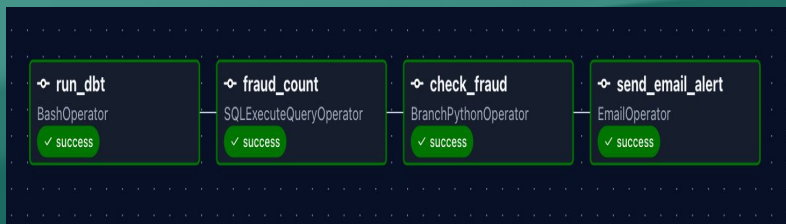
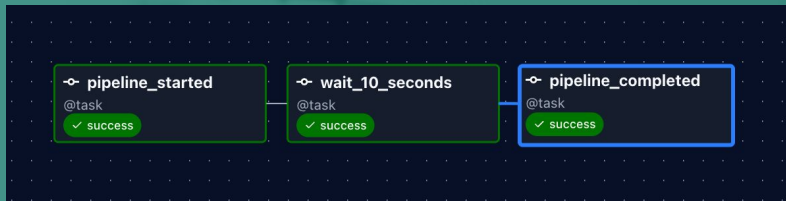
### Advanced Pandas Assessment 🐼

Objective: Solve the following advanced-level Python Pandas and Data Analytics exercises using the Telecom Customer Churn dataset.

Module 1: Data Profiling & Quality

Module 2: Advanced Customer Analytics

# Phase 3



## Airflow, dbt and Snowflake

**DAG 1 - Create a DAG that runs daily at 8:00 AM**

*Create a DAG that runs daily at 8:00 AM.*

**DAG 2 - Load and transform data in Snowflake.**

*Load and transform data in Snowflake.*

**DAG 3 - Banking Fraud Detection Project**

*Build a fraud analytics pipeline.*

# Fraud alert notification

Connect Airflow to smtp



Fraud Alert

Inbox x



moralessalcedoandrea@gmail.com

to me ▾

## Fraud Alert Triggered

Suspicious transactions have been detected.

Please check the fraud\_alerts table in Snowflake.

Host: smtp.gmail.com

Port: 587

Username: Gmail address

Password: Gmail App Password

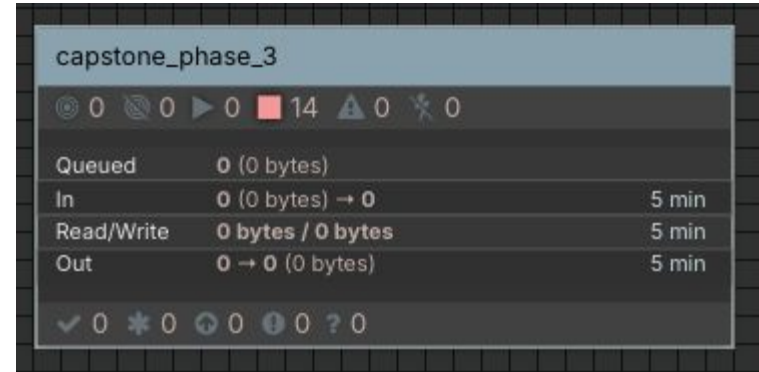
Disable SSL: ☒ Yes

Disable TLS: ☐ No

# Phase 4

Nifi

Kafka



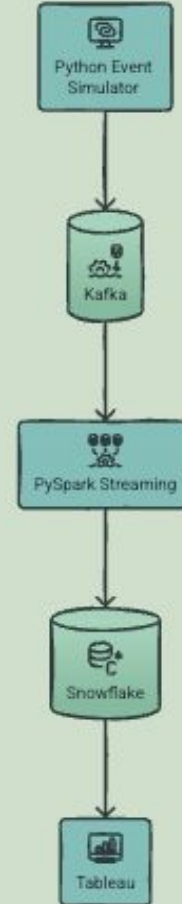
- NiFi flow to read CSV files >> filename format: row\_.txt.
- NiFi dataflow that polls a REST API
- Ingest data from a Kafka
- Python Kafka producer that reads rows from a CSV file
- Kafka Streams application

## Phase 5

Build a real-time social media analytics platform

**Technology Stack:** Python Event Simulator, Kafka, PySpark Streaming, Snowflake and Tableau.

### Social Media Data Processing Pipeline



---

# Event simulator 🐍



```
social_media_analytics > simulator > event_simulator.py > ...
1  import json
2  import random
3  import time
4  import uuid
5  from datetime import datetime, timezone, timedelta
6
7  from kafka import KafkaProducer
8
9
10 KAFKA_BOOTSTRAP_SERVERS = "localhost:9092"
11 KAFKA_TOPIC = "social_media_events"
12
13 NUMBER_OF_USERS = 100
14 NUMBER_OF_POSTS = 500
15
16 # 20 events/second = 1,200 events/minute
17 EVENTS_PER_SECOND = 20
18
19 # Post that will be used for viral-content detection
20 VIRAL_POST_ID = "post_1"
21
22
23 COUNTRIES = [
24     "Netherlands",
25     "Germany",
26     "Spain",
27     "France",
28     "United Kingdom",
29     "Italy",
30     "Belgium",
31     "Portugal",
32     "United States",
33     "Canada"
34 ]
35
36 HASHTAGS = [
```



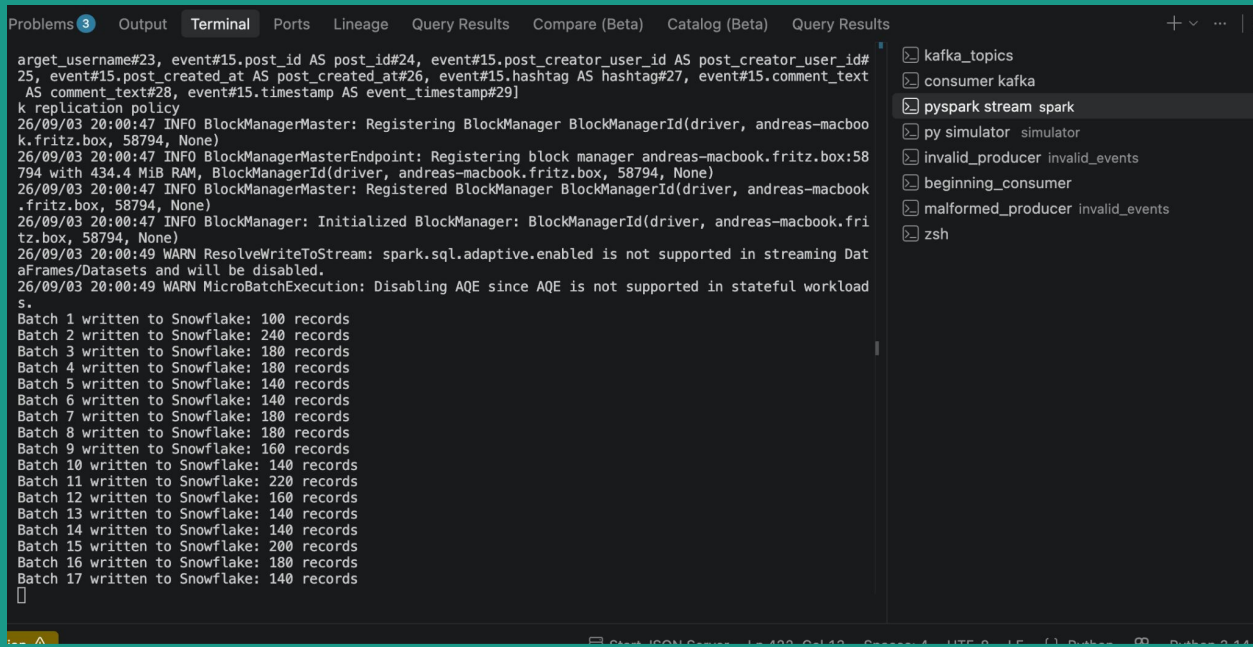
# Kafka ingestion layer

```
Problems Output Terminal Ports Lineage Query Results Compare (Beta) Catalog (Beta) Query Results + v ... | {} X

{"event_id": "e6522b43-4e8a-4acb-b48a-62509da6f4fc", "event_type": "like", "user_id": "user_83", "username": "user_083", "country": "Italy",
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st_creator_user_id": "user_26", "post_created_at": "2026-05-25T23:57:56.057262+00:00", "hashtag": "#music", "comment_text": null, "timestamp"
: "2026-09-03T17:55:38.156556+00:00"}
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70", "post_creator_user_id": "user_97", "post_created_at": "2026-02-14T09:51:51.056803+00:00", "hashtag": "#music", "comment_text": null, "ti
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d Kingdom", "user_created_at": "2026-08-14T17:32:12.055633+00:00", "target_user_id": "user_89", "target_username": "user_089", "post_id": "po
st_11", "post_creator_user_id": "user_89", "post_created_at": "2026-04-25T04:19:38.055823+00:00", "hashtag": "#food", "comment_text": "Intere
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ost_creator_user_id": "user_11", "post_created_at": "2026-03-25T04:11:36.057050+00:00", "hashtag": "#fitness", "comment_text": null, "timest
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ost_403", "post_creator_user_id": "user_56", "post_created_at": "2025-12-19T13:48:26.057294+00:00", "hashtag": "#fitness", "comment_text": n
ull, "timestamp": "2026-09-03T17:55:38.157617+00:00"}
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aly", "user_created_at": "2026-03-06T23:45:34.055569+00:00", "target_user_id": "user_29", "target_username": "user_029", "post_id": "post_285
", "post_creator_user_id": "user_29", "post_created_at": "2025-09-12T12:42:57.056866+00:00", "hashtag": "#fitness", "comment_text": null, "ti
mestamp": "2026-09-03T17:55:38.157827+00:00"}

```

# Real-Time Processing with PySpark



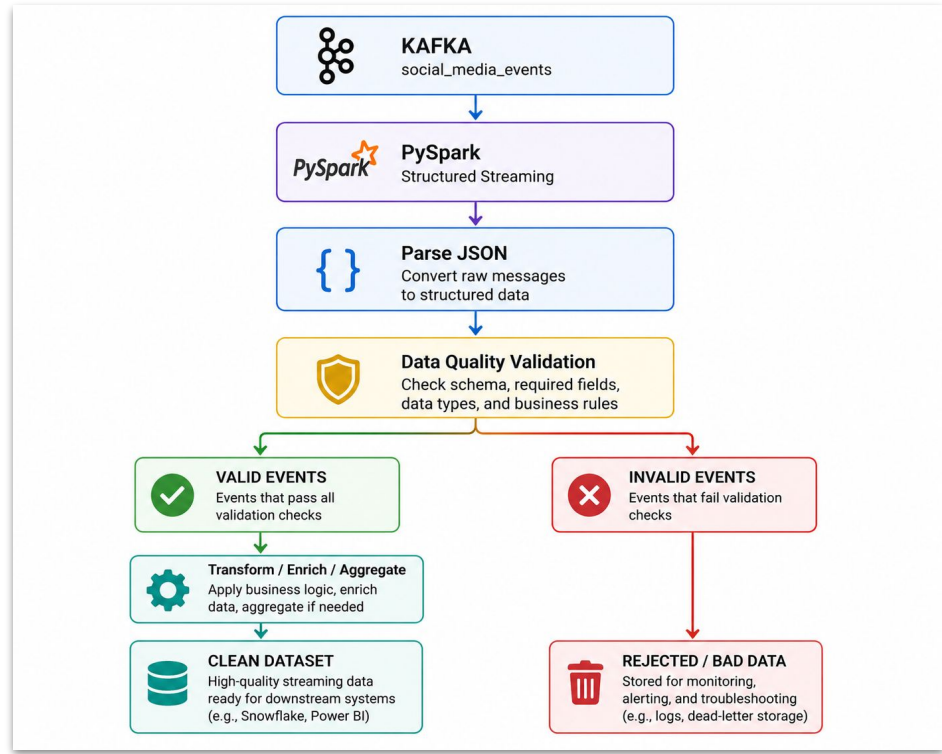
The screenshot shows a PySpark application running in a terminal window. The terminal output displays logs for the BlockManagerMaster and BlockManagerMasterEndpoint, indicating the registration of the driver and the initialization of the BlockManager. It also shows warnings about unsupported features like ResolveWriteToStream and MicroBatchExecution. The application is writing data to Snowflake in batches, with the following records per batch:

- Batch 1 written to Snowflake: 100 records
- Batch 2 written to Snowflake: 240 records
- Batch 3 written to Snowflake: 180 records
- Batch 4 written to Snowflake: 180 records
- Batch 5 written to Snowflake: 140 records
- Batch 6 written to Snowflake: 140 records
- Batch 7 written to Snowflake: 180 records
- Batch 8 written to Snowflake: 180 records
- Batch 9 written to Snowflake: 160 records
- Batch 10 written to Snowflake: 140 records
- Batch 11 written to Snowflake: 220 records
- Batch 12 written to Snowflake: 160 records
- Batch 13 written to Snowflake: 140 records
- Batch 14 written to Snowflake: 140 records
- Batch 15 written to Snowflake: 200 records
- Batch 16 written to Snowflake: 180 records
- Batch 17 written to Snowflake: 140 records

The file explorer on the right side of the terminal window shows the following files and directories:

- kafka\_topics
- consumer kafka
- pyspark stream spark
- py simulator simulator
- invalid\_producer invalid\_events
- beginning\_consumer
- malformed\_producer invalid\_events
- zsh

# Invalid and/or malformed events



# Data Enrichment

--- overview

```
SELECT *
FROM SOCIAL_MEDIA_DB_AMS.CURATED.CURATED_EVENTS
ORDER BY EVENT_TIMESTAMP DESC
LIMIT 10;
```

--- step 7 Trending Hashtag Analytic

--- Create TRENDING\_HASHTAGS

```
CREATE TABLE IF NOT EXISTS SOCIAL_MEDIA_DB_AMS.ANALYTICS.TRENDING_HASHTAGS (
  WINDOW_MINUTES INTEGER,
  HASHTAG STRING,
  EVENT_COUNT INTEGER,
  RANK INTEGER,
  WINDOW_START TIMESTAMP_TZ,
  WINDOW_END TIMESTAMP_TZ
);
```

SOCIAL\_MEDIA\_DB\_AMS / CURATED / CURATED\_EVENTS

Table

ACCOUNTADMIN

3 minutes ago

9.9K

590.0KB

Table Details

Columns

Data Preview

Copy History

Data Quality

Lineage

+ COMPUTE\_WH 100 of 9.9K Rows • Updated just now

|    | EVENT_ID                             | EVENT_TYPE    | USER_ID | USERNAME | COUNTRY        | TARGET_USER_ID | TARGET_USERNAME |
|----|--------------------------------------|---------------|---------|----------|----------------|----------------|-----------------|
| 1  | 01bc8ddc-ef86-4038-a27a-ef527f56e7e0 | comment       | user_10 | user_010 | Netherlands    | user_23        | user_023        |
| 2  | 298616f9-d396-4869-a34d-6d8ab56e1cf1 | profile_visit | user_60 | user_060 | Canada         | user_72        | user_072        |
| 3  | ab55c091-f840-4634-9674-05da5dec7ad6 | like          | user_39 | user_039 | Belgium        | user_73        | user_073        |
| 4  | 17b448c-3132-4982-8195-aef12d5864d7  | video_view    | user_9  | user_009 | Spain          | user_35        | user_035        |
| 5  | 2383f162-59b6-45a8-a07b-3c83c35b6527 | like          | user_67 | user_067 | Italy          | user_33        | user_033        |
| 6  | 2b7ef880-8ba6-434e-a4e6-d7adba344ba1 | like          | user_11 | user_011 | Germany        | user_28        | user_028        |
| 7  | ba73b6c6-ad06-4acf-a172-27229e5320e8 | profile_visit | user_27 | user_027 | Belgium        | user_21        | user_021        |
| 8  | fa53b4f8-7603-4e43-bdde-ba668c0cdbc8 | like          | user_16 | user_016 | Italy          | user_42        | user_042        |
| 9  | 5248d3c7-2edf-4cdd-ba7a-47785a48f467 | comment       | user_71 | user_071 | Netherlands    | user_38        | user_038        |
| 10 | c6848f2b-014b-4cf1-a851-b147d2bda2d2 | comment       | user_4  | user_004 | Netherlands    | user_5         | user_005        |
| 11 | db896663-34cc-45a8-ac2a-becb1a487f8e | comment       | user_8  | user_008 | Portugal       | user_16        | user_016        |
| 12 | 0ca25e17-44b5-429a-bfad-f8c24083a04c | video_view    | user_22 | user_022 | Italy          | user_26        | user_026        |
| 13 | fb93374a-0992-4bea-b229-28873407ef19 | like          | user_16 | user_016 | Italy          | user_39        | user_039        |
| 14 | 489f251f-9991-4daa-8946-9770cacabb0c | comment       | user_41 | user_041 | Germany        | user_3         | user_003        |
| 15 | 52e156d7-1d66-421a-8c91-18ed58ea3218 | like          | user_84 | user_084 | Canada         | user_74        | user_074        |
| 16 | 2b8795d3-bcc5-4fe7-8b3f-301132e4e60b | like          | user_82 | user_082 | France         | user_36        | user_036        |
| 17 | e45af1c4-2d97-42c8-b298-6406039ce05e | like          | user_67 | user_067 | Italy          | user_14        | user_014        |
| 18 | 5d106ca8-bd30-466f-b159-29f844314dce | share         | user_46 | user_046 | France         | user_28        | user_028        |
| 19 | 8e107479-fef1-4693-a664-g2da28ee405  | like          | user_44 | user_044 | France         | user_81        | user_081        |
| 20 | 247325c4-418b-4dab-a9ee-f0f2bfba8b20 | like          | user_48 | user_048 | United Kingdom | user_81        | user_081        |
| 21 | 6a5ef98f-b7ba-44d4-b41f-17881bcb289a | like          | user_87 | user_087 | Netherlands    | user_81        | user_081        |
| 22 | 246c6569-2f80-4a19-9c58-d04c436fd998 | like          | user_83 | user_083 | Italy          | user_35        | user_035        |
| 23 | 78d15a1e-15f3-4f20-ac23-b170ac661f26 | share         | user_84 | user_084 | Canada         | user_2         | user_002        |
| 24 | 1c062033-0d5e-49b4-b3d5-816488e605da | like          | user_89 | user_089 | Germany        | user_36        | user_036        |
| 25 | 4ef3d54f-8a8d-4257-9fb0-6bb5a338dc48 | video_view    | user_4  | user_004 | Netherlands    | user_33        | user_033        |
| 26 | 657c39c7-c778-47b2-af9b-c8f7f9f7b51f | share         | user_9  | user_009 | Spain          | user_38        | user_038        |
| 27 | 244fa98b-a334-4993-be30-6bae9767ad10 | video_view    | user_85 | user_085 | United States  | user_84        | user_084        |
| 28 | e8dfcd08-4889-40a3-9b0d-baab4801678  | profile_visit | user_17 | user_017 | Portugal       | user_48        | user_048        |
| 29 | 6a7d1139-f58b-4730-87fd-cd258bd78662 | like          | user_3  | user_003 | Netherlands    | user_73        | user_073        |
| 30 | e5ddfb8e-5e78-4a73-97ed-fd06bfb1baac | video_view    | user_50 | user_050 | Germany        | user_77        | user_077        |
| 31 | b924a8d0-970a-4856-85d2-493240e8a656 | like          | user_38 | user_038 | Belgium        | user_9         | user_009        |
| 32 | 905466ce-a82f-400c-9577-73464a9a6810 | share         | user_14 | user_014 | United States  | user_69        | user_069        |

# Dashboards



### Total Engagement

28,802

### Total Events

52,340

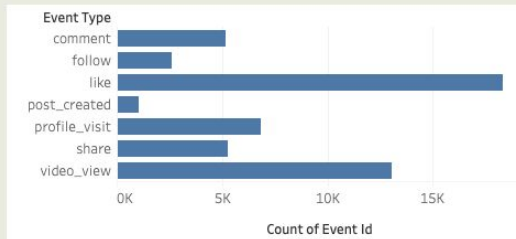
### Active Users

52,340

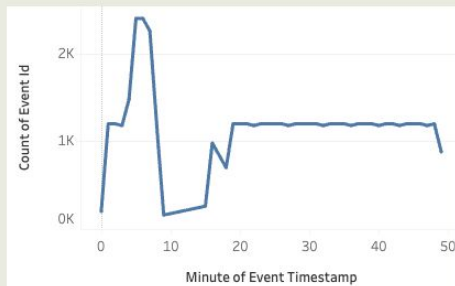
### Video Views

| Event Type |        |
|------------|--------|
| video_view | 13,029 |

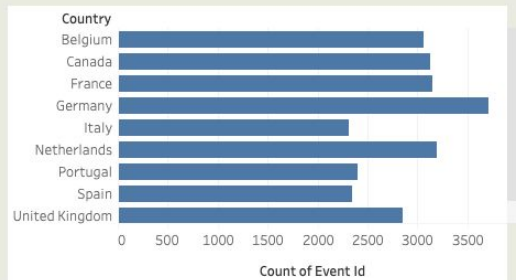
### Engagement by Event Type



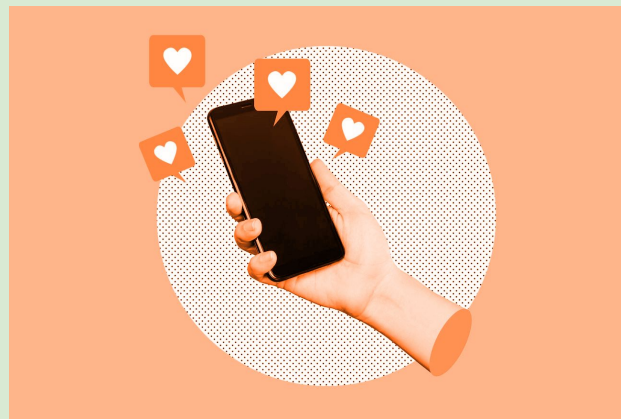
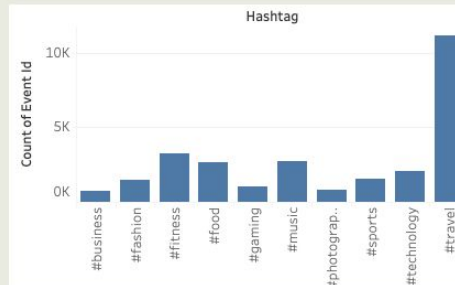
### Events Over Time



### Engagement by Country



### Top Hashtags by Engagement





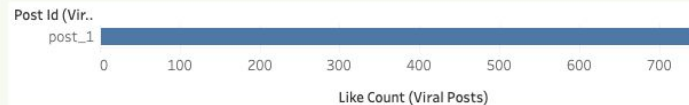
### KPI Viral Posts

1

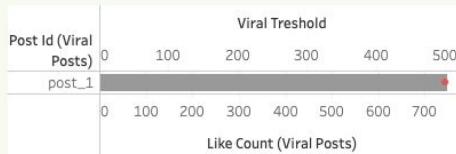
### Highest Likes KPI

747

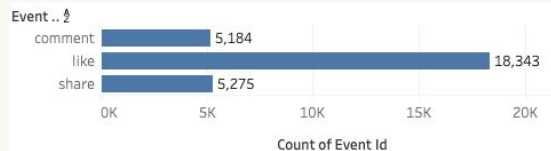
### Likes per Viral Post



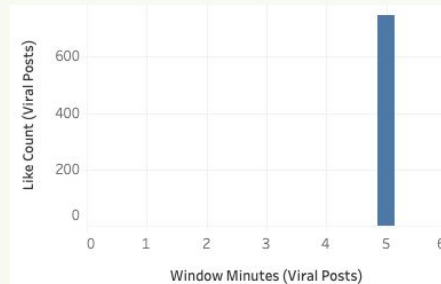
### Viral Threshold



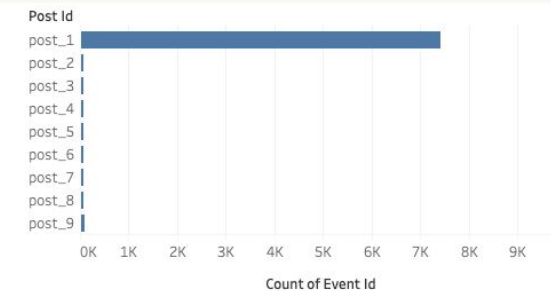
### What makes a post viral?



### Viral Detection Over Time



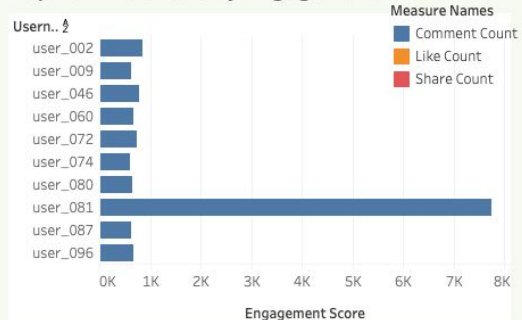
### Top 10 Posts by Likes



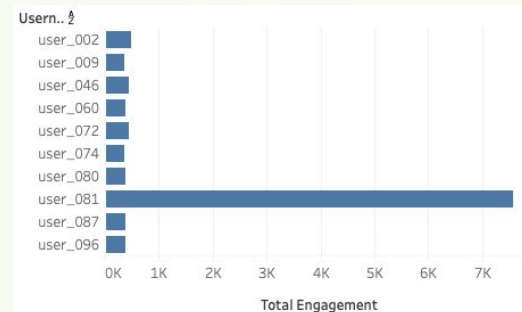


## Influencer Dashboard

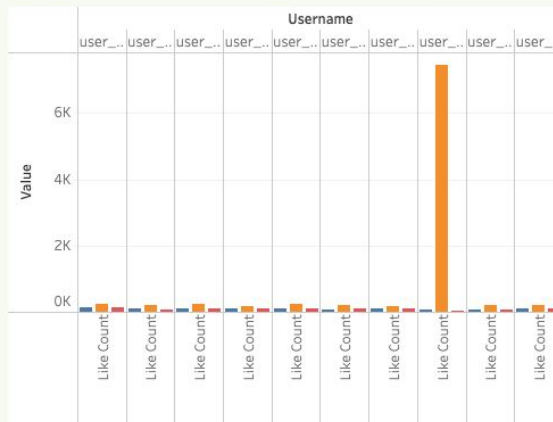
### Top 10 Influencers by Engagement Score



### Total Engagement by Influencer



## Likes, Comments & Shares by Influencer



### Influencer Ranking Table

| Username | Engage.. | Like Co.. | Rank | Share C.. | Total E.. |
|----------|----------|-----------|------|-----------|-----------|
| user_081 | 7,738    | 7,461     | 1    | 51        | 7,574     |
| user_002 | 860      | 248       | 2    | 122       | 493       |
| user_046 | 790      | 230       | 3    | 116       | 452       |
| user_072 | 745      | 235       | 4    | 108       | 436       |
| user_060 | 682      | 178       | 5    | 104       | 378       |
| user_096 | 673      | 197       | 6    | 90        | 390       |
| user_080 | 662      | 187       | 7    | 93        | 378       |
| user_087 | 628      | 206       | 8    | 84        | 375       |
| user_009 | 619      | 197       | 9    | 82        | 367       |
| user_074 | 608      | 192       | 10   | 86        | 357       |
| user_005 | 599      | 200       | 11   | 85        | 357       |
| user_032 | 597      | 171       | 12   | 80        | 344       |
| user_016 | 583      | 164       | 13   | 81        | 333       |







Thank you!